

Module 29 - Generator Functions

1. What is a Generator Function?

A Generator Function is a special type of function that can pause its execution using the `yield` keyword and continue from the same point when `next()` is called. It is declared using `function*`.

Example Code 1:

```
// Generator Function
// 'function*' declares a generator function.
function* generatorFunction() {
    // Pauses the function and returns "First Val".
    yield "First Val";
    // Continues from the previous pause and returns "Second Val".
    yield "Second Val";
    // Continues again and returns "Third Val".
    yield "Third Val";
    // Ends the generator and returns the final value.
    return "Final Val";
}
// Creates a generator object.
let generator = generatorFunction();
// 'next()' resumes the function until the next 'yield' or 'return'.
console.log(generator.next().value);
console.log("Im executing after first yield statement");
console.log(generator.next().value);
console.log(generator.next().value);
console.log(generator.next().value);
```

Example Code 2:

```
function* url() {  
    // Returns each part of the URL one by one.  
    yield "https://";  
    yield "www.uniqtech.com";  
    yield "homePage";  
}  
// Creates a generator object.  
let origin1 = url();  
// Each 'next()' returns the next yielded value.  
console.log(origin1.next().value);  
console.log(origin1.next().value);  
console.log(origin1.next().value);  
// No more values are available, so it returns undefined.  
console.log(origin1.next().value);
```